

S Madhan Kumar

Data Analyst

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EXPERIENCE

Data Science Intern

Jan 2025 - Feb 2025

EdiGlobe

- **Engineered end-to-end analytical pipelines** to solve complex business bottlenecks, directly contributing to the achievement of quarterly strategic goals.
- **Optimized operational workflows** by leveraging data insights, resulting in measurable enhancements to business value and process efficiency.
- **Synthesized critical business intelligence** for high-level stakeholders, facilitating data-driven decision-making through advanced statistical modeling.
- **Developed interactive data visualizations** and predictive models to interpret trends and influence strategic company-wide initiatives.

EDUCATION

Vel Tech University

2023 -

Bachelor's of Technology in Computer Science (Data Science) • GPA: 8.84

2027

Patent Pending: Hybrid SOC Estimation Framework utilizing IoT & Satellite Data (Google Earth Engine) for high-precision environmental analytics.

Everwin Vidhyasharam

2022 - 2023

12th in MPCB • GPA: 8.18

PROJECTS

Multi-Modal MRV Framework [\[Link\]](#)

2025 - Present

Founder & Lead Developer

[Case Study / Architecture Documentation Available Upon Request, Repository is private]

- Developed a patent-pending hybrid framework for Soil Organic Carbon (SOC) estimation using IoT and Satellite data.
- Fused Sentinel-2 multi-spectral imagery with real-time ground telemetry from ESP32 sensors for high-precision environmental analytics.
- Automated data ingestion pipelines to process atmospheric Vapor Pressure Deficit (VPD) and soil metrics.

Ancient Tamil Character Recognition [\[Link\]](#)

2024-2025

Lead Developer

- Built an AI-powered web application using TensorFlow/Keras and FastAPI to decipher 8th-century Vatteluthu inscriptions.
- Developed a CNN-based OCR model to translate ancient scripts into modern Tamil, preserving historical linguistic data.

NL-to-SQL Intent Engine [\[Link\]](#)

2024 - 2025

Lead Developer

- Architected a modular NLP bot using Scikit-learn for intent-based classification, translating natural language into executable SQL for relational datasets.

- Deployed a high-performance stack featuring a FastAPI backend and a Streamlit frontend to reduce data access friction for non-technical users.

SKILLS

Technical Skills

- Programming: Python (Advanced), Java (Spring Boot), SQL (Recursive Joins/Aggregates), C, HTML/JS.
- Data & AI: TensorFlow, Keras, FastAPI, Scikit-learn, CNN Architecture, Data Pipelines.
- Tools: Git, LM Studio, OpenClaw, Aider, Unity, Streamlit, Bootstrap 5.
- Databases: MySQL (Hierarchical Schema Design, Multi-table Handshakes).
- Specialized: Google Earth Engine (Satellite Data), Sentinel-2 Analysis, ESP32 Telemetry.

Soft Skills

- Leadership: Founder/Lead Dev of AegisLM; managing end-to-end project lifecycles.
- Strategy: Strategic Problem Solving; translating business goals into technical solutions.
- Communication: Data-driven storytelling for high-level stakeholder decision-making.

Languages

- English & Tamil: Professional Proficiency / Native.
- German: Elementary (A1); currently attending weekend A2 certification classes.